

Applying the LULC classifier on real sites: robustness, data adequacy, and validation

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M.Tech CSE Entry No. 2025MCS2106

July 2026

Where we are, and what was advised

Done so far

- A 4-class base map over India at 10 m, with a living hierarchy: split or add a class, give a few example polygons, and retrain that node on the fly.
- A git-backed zoo of Model and Dataset cards, with save and reload, merging, base and year selection, and publishing to a shared repository.

What was advised

- Start applying the framework on real sites: acacia against non-acacia, tea against non-tea, and mining.
- Make the models robust across years by training on more than one year.
- Judge training data by how much of the area it covers and how it is spread, not by a raw count.
- Validate an uploaded scheme before running it, and revisit the data choices: inference year and Tessera.

Standard testing sites

- We fixed a home strip around IIT Delhi and Sanjay Van: built-up, tree cover, and a small water body in one small area, with acacia and non-acacia crowns inside it.
- Three such sites are now default areas in the tool, chosen from a menu without typing coordinates: IIT Delhi with Sanjay Van, Asola Bhatti, and Jalpaiguri, alongside the Assam tea belt already present.
- For acacia, we now hold 912 confidently labelled tree crowns, 336 acacia and 576 non-acacia, prepared as ready training examples for the tool.
- Asola Bhatti serves as a false-positive probe: old mines there were reclaimed into built-up and scrub, so a mining model should stay quiet and an acacia model may not.

Robustness across years

- Vegetation changes from year to year, and Alpha Earth has a separate yearly image, so a model trained on one year can drift on another.
- A split can now be trained on several years at once. The same polygon sampled in different years becomes more training examples, while a whole polygon is still held out together, so the test stays honest.
- On acacia and non-acacia, training on 2019, 2021, and 2023 and testing on the unseen 2020 and 2024 raised accuracy from 0.635 to 0.745, a gain of about eleven points over a single-year model.
- On the four base classes the multi-year and single-year models are the same, both near 0.89 on the unseen years. The coarse classes look alike year to year, so a single year already generalises; the gain from multiple years shows up on the finer, year-sensitive splits.

Testing on the sites: what we found

- Tea against non-tea separates cleanly. Acacia against non-acacia is genuinely hard, since it is a species distinction, but it is where training on several years helps most.
- The mining model detects real mines well: an active coalfield (Jharia) reads 71% mining. On reclaimed, mine-like ground it raises false positives: reclaimed Asola reads 17%, which is exactly the concern raised for Asola Bhatti. Same detector on both, so the two are comparable.

Test	Result
Tea vs non-tea	0.957 held-out accuracy
Acacia vs non-acacia	0.745 on unseen years (multi-year); a hard split
Mining detection	0.859 accuracy, 0.854 recall on real mines
Mining false positives	0.139 on non-mining ground truth
Jharia, active coalfield	71.2% of the area flagged mining (real detections)
Asola Bhatti, reclaimed	17.1% flagged mining (mostly false positives)

Judging whether there is enough training data

- A fixed number of labelled points means little on its own: the same count is plenty over a village and almost nothing over a district.
- We now report coverage: how much of the area about to be classified actually falls inside a labelled polygon. It is read against the size of that area, so it rises on a small area and falls on a large one.
- Alongside it we keep the spread measure, the spatial evenness of the labels across the region, since data clustered in one place skews a model.

Validating a scheme before it runs

- A user can upload a saved scheme: the class hierarchy and the ordered steps that built it. Until now it was applied first and any problem was noticed afterwards.
- There is now a check that runs before anything is applied and changes nothing. It reports errors that block loading and warnings that do not.
- It confirms the hierarchy is well formed, that every recorded operation is one the tool knows and carries its required fields, and that each classifier the scheme refers to is present.
- Loading now runs this check first, so a broken file is refused before it can alter the live tree.

Choosing the data: years and Tessera

- The inference-year picker now carries a note on how far a ground truth stays reliable across years, so the user knows what a chosen year buys.
- Tessera remains available as the second feature source in Detailed mode. It has usable India coverage only for 2024, and the note states that other years can be requested.

Thank you